

Task 1: Install, Configure, and Test Snort Intrusion Detection System (IDS)

Objective

The objective of this task was to install, configure, and test Snort as a Network Intrusion Detection System (NIDS) on Kali Linux. Custom detection rules were created to monitor ICMP (ping), TCP SYN (Nmap scan), and HTTP traffic. Metasploitable 2 was used to generate network traffic, while Snort monitored and generated alerts.

Lab Environment

Component	Description
Operating System	Kali Linux
IDS Software	Snort 3.12.2.0
Target Machine	Metasploitable 2
Kali IP Address	10.0.2.15
Metasploitable IP Address	10.0.2.4
Network Interface	eth0

Step 1: Verify Snort Installation

Snort was installed on Kali Linux and tested using the following command:

```
sudo snort -T -c /etc/snort/snort.lua
```

The output confirmed:

Snort successfully validated the configuration (with 0 warnings).

Insert Screenshot 1: *Snort configuration validation.*

Step 2: Create Custom Detection Rules

The custom rules were added to:

```
/etc/snort/rules/local.rules
```

Rule 1: ICMP Detection

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Purpose: Detects ICMP Echo Request and Echo Reply packets generated during a ping.

Rule 2: Nmap Scan Detection

```
alert tcp any any -> any any (msg:"Nmap Scan Detected"; flags:S; sid:1000002; rev:1;)
```

Purpose: Detects TCP SYN packets commonly generated during an Nmap SYN scan.

Rule 3: HTTP Traffic Detection

```
alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000003; rev:1;)
```

Purpose: Detects HTTP traffic on TCP port 80.

Insert Screenshot 2: *Contents of local.rules.*

Step 3: Start Snort

Snort was started using the following command:

```
sudo snort -q -A alert_fast -i eth0 -c /etc/snort/snort.lua -R  
/etc/snort/rules/local.rules
```

The command loaded all custom rules and started monitoring network traffic on interface **eth0**.

Insert Screenshot 3: *Snort running successfully.*

Step 4: Testing the ICMP Rule

A ping test was performed from Metasploitable to Kali:

```
ping -c 4 10.0.2.15
```

Snort generated the following alert:

```
[1:1000001:1] ICMP Ping Detected
```

Analysis

The ICMP rule successfully detected ping traffic between the two virtual machines. This type of detection is useful because attackers often use ping sweeps to identify active hosts before launching further attacks.

Insert Screenshot 4: *ICMP Ping Detected alert.*

Step 5: Testing the Nmap Rule

An Nmap scan was executed:

```
nmap 10.0.2.15
```

Snort generated multiple alerts:

```
[1:1000002:1] Nmap Scan Detected
```

Analysis

The rule successfully detected TCP SYN packets generated during the Nmap scan. Detecting reconnaissance activities such as port scanning enables security teams to identify attackers before exploitation attempts begin.

Insert Screenshot 5: *Nmap Scan Detected alert.*

Step 6: Testing the HTTP Rule

HTTP traffic was generated by scanning port 80:

```
nmap -p 80 10.0.2.4
```

Snort generated the alert:

```
[1:1000003:1] HTTP Traffic Detected
```

Analysis

The rule successfully detected HTTP traffic destined for TCP port 80. Monitoring HTTP traffic helps identify suspicious web requests, unauthorized access attempts, and possible web-based attacks.

Insert Screenshot 6: *HTTP Traffic Detected alert.*

Task 4: Analysis of Snort Alerts

ICMP Ping Detected

- **Traffic Type:** ICMP Echo Request and Echo Reply
 - **Purpose:** Detects ping activity.
 - **Security Significance:** Attackers often use ping to identify active systems before conducting further reconnaissance or attacks.
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Nmap Scan Detected

- **Traffic Type:** TCP SYN packets
 - **Purpose:** Detects port scanning activity.
 - **Security Significance:** Port scanning is a common reconnaissance technique used to discover open ports and vulnerable services.
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HTTP Traffic Detected

- **Traffic Type:** TCP Port 80
 - **Purpose:** Detects HTTP communication.
 - **Security Significance:** HTTP monitoring helps identify suspicious web requests, unauthorized access attempts, and web-based attacks.
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Conclusion

The objective of this practical exercise was successfully achieved. Snort was installed, configured, and tested on Kali Linux as a Network Intrusion Detection System. Three custom detection rules were implemented to monitor ICMP, TCP SYN (Nmap), and HTTP traffic. Testing with Metasploitable 2 confirmed that Snort successfully detected each type of network activity and generated alerts in real time. This demonstrates how Snort can be used to monitor network traffic, identify reconnaissance activities, and improve network security by providing early warning of suspicious events.

This report is based on the exact steps you completed in your lab, so it matches the screenshots you captured. It should fit well with the assignment requirements.